

SQL Practical Assignment

Constraints & Index

Scenario: College Student Management System

Create a database and tables for a college student management system. Perform all tasks practically in MySQL.

Part 1 – Database

Q1. Create a database named **college_db** and select it using **USE**.

Part 2 – Departments Table

Q2. Create a **departments** table with the following requirements:

Column	Requirement
dept_id	PRIMARY KEY
dept_name	NOT NULL + UNIQUE

Q3. Insert these four departments: Computer, Management, Commerce and Science.

Part 3 – Students Table

Q4. Create a **students** table using all of the following:

Column	Requirement
student_id	PRIMARY KEY + AUTO_INCREMENT
student_name	NOT NULL
age	CHECK (age >= 18)
city	DEFAULT 'Pune'
email	UNIQUE
gender	ENUM('Male','Female','Other')
dept_id	FOREIGN KEY → departments(dept_id)

Part 4 – Insert Records

Q5. Insert at least **10 students**. Do not manually enter student_id.

Q6. Make sure at least 2 students are from Pune, 2 from Mumbai and 2 from other cities.

Q7. Insert students belonging to at least 3 different departments.

Q8. Use all three ENUM values: Male, Female and Other.

Q9. For at least one student, do not provide the city and verify that Pune is inserted automatically.

Part 5 – Practical Constraint Testing

Q10. Try inserting a student with **NULL** as student_name. Observe the result.

Q11. Try inserting a student with age **15**. Observe the CHECK constraint.

Q12. Try inserting two students with the same email. Observe the UNIQUE constraint.

Q13. Try inserting a student with gender '**Unknown**'. Observe the ENUM restriction.

Q14. Try inserting a student with a dept_id that does not exist in departments. Observe the FOREIGN KEY restriction.

Part 6 – Index

Q15. Create an index named **idx_city** on the **city** column.

Q16. Display the indexes of the students table using **SHOW INDEX**.

Q17. Search and display all students whose city is Pune.

Part 7 – Practical SQL Queries

Q18. Display all records from students.

Q19. Display only student_name, email and city.

Q20. Display students whose age is greater than 20.

Q21. Display students whose age is between 18 and 25.

Q22. Display all female students.

Q23. Display students from Mumbai.

Q24. Display students belonging to the Computer department using JOIN.

Q25. Display student name, city and department name using JOIN.

Q26. Display students whose department is Management or Commerce using JOIN.

Q27. Display all students in ascending order of age.

Q28. Display the student with the highest age.

Practical Submission

Submit all **CREATE**, **INSERT**, constraint-testing, **INDEX** and **SELECT**/JOIN queries, along with screenshots showing successful execution and constraint errors.

Total: 50 Marks | Database & Tables: 10 | Data & Constraints: 15 | Queries & JOIN: 20 | Index: 5